

Mechanistic Interpretability of Gen Z Slang in Large Language Models

Overview

This project explores whether a Large Language Model (LLM) contains an internal representation corresponding to **Gen Z slang and internet-speak style**, and whether this representation can be manipulated using activation steering techniques.

Using the **Gemma-2-2B-Instruct** model and the **TransformerLens** library, we identify a direction in activation space that separates responses written in Gen Z slang from responses written in formal English. We then use this direction to:

1. Increase slang usage in neutral responses.
2. Suppress slang usage in slang-oriented prompts.
3. Analyze where stylistic information is encoded inside the model.

The project demonstrates a practical application of **mechanistic interpretability** and **activation steering** for controlling linguistic style without fine-tuning model weights.

Objectives

The main goals of this project were:

- Identify an internal activation direction associated with Gen Z slang.
 - Locate layers that strongly encode stylistic information.
 - Verify whether this direction can steer model outputs.
 - Demonstrate both style injection and style suppression through activation interventions.
-

Methodology

1. Model Selection

The project uses:

- Model: Gemma-2-2B-Instruct
- Framework: TransformerLens
- Hardware: GPU (CUDA when available)

The model is loaded using:

```
HookedTransformer.from_pretrained_no_processing(  
    "google/gemma-2-2b-it"  
)
```

2. Dataset Construction

To isolate style from content, paired prompts were created.

Example:

Slang Prompt

```
Explain climate change. Respond using Gen Z slang and internet speak.
```

Plain Prompt

```
Explain climate change. Use formal, plain English with no slang.
```

The topic remains identical while only the writing style changes.

A collection of topics was used, including:

- Climate Change
- Artificial Intelligence
- DNA and Genetics
- Quantum Computing
- The Internet
- Photosynthesis
- Renewable Energy
- Black Holes

This ensures the learned direction represents style rather than subject matter.

3. Activation Extraction

For every prompt, hidden activations were extracted from:

```
resid_pre
```

across all transformer layers.

Only the final semantic position (`pos = -1`) was used.

For each layer:

μ_{slang}

and

μ_{plain}

were computed.

4. Computing the Slang Direction

The steering direction was defined as:

$\text{Direction} = \mu_{\text{slang}} - \mu_{\text{plain}}$

and normalized:

$d = \text{Direction} / ||\text{Direction}||$

This vector represents the dominant activation pattern associated with Gen Z style.

5. Layer Analysis

Layer-wise separation was measured using:

$||\mu_{\text{slang}} - \mu_{\text{plain}}||$

The layer with the highest separation was selected as the most style-sensitive layer.

This identifies where the model most strongly distinguishes between formal and slang writing styles.

6. PCA Visualization

Principal Component Analysis (PCA) was applied to activations from the most separable layers.

Observations:

- Slang activations formed a distinct cluster.
 - Plain-English activations formed another cluster.
 - Clear separation indicated that style information is encoded geometrically within activation space.
-

7. Activation Steering

A forward hook was applied to inject the slang direction during generation.

Steering operation:

$$h' = h + \alpha d$$

where:

- h = original activation
- d = slang direction
- α = steering coefficient

The intervention was applied to:

hook_resid_pre

across all transformer layers.

Experiments

Experiment 1: Slang Injection

Neutral prompts were used:

Tell me about the ocean.
Explain how rainbows form.
What is thunder?
Describe the human heart.

Without steering:

- Responses remained neutral and educational.

With steering:

- Responses increasingly adopted internet slang.
- Terms such as:
 - slay
 - lowkey
 - vibe
 - no cap
 - iconic
 - based

appeared naturally in generated text.

Experiment 2: Coefficient Sweep

Steering coefficients tested:

0, 5, 10, 15, 20, 25, 30, 40

Results showed:

- Larger coefficients increased slang frequency.
- Small coefficients produced subtle stylistic shifts.
- Very large coefficients occasionally reduced fluency.

This demonstrates a controllable relationship between steering strength and stylistic behavior.

Experiment 3: Slang Ablation

To remove stylistic behavior, the projection of activations onto the slang direction was subtracted.

Ablation:

$$h' = h - \text{proj}_d(h)$$

Results:

- Prompts requesting Gen Z slang produced noticeably fewer slang terms.
- Responses became more neutral and formal.

This suggests the identified direction is causally linked to style generation.

Key Findings

1. Style Has a Dedicated Representation

Gen Z slang is not distributed randomly throughout the network.

A consistent activation direction captures a significant portion of stylistic behavior.

2. Activation Steering Works

Adding the direction:

- Increases slang usage.

Removing the direction:

- Suppresses slang usage.

This provides evidence of a causal relationship.

3. Style Information Is Layer-Specific

Certain transformer layers exhibit much stronger separation between slang and plain responses.

This suggests stylistic features emerge progressively through the network.

4. No Fine-Tuning Required

The model's weights remain unchanged.

Behavior is controlled entirely through activation manipulation during inference.

Applications

This technique can be extended to:

- Formal ↔ Informal writing
- Professional communication
- Toxicity reduction
- Sentiment steering
- Humor control
- Safety interventions

- Persona and character control
-

Technologies Used

- Python
 - PyTorch
 - TransformerLens
 - NumPy
 - Scikit-Learn
 - Plotly
 - Hugging Face Transformers
 - Gemma-2-2B-Instruct
-

Conclusion

This project demonstrates that Gen Z slang corresponds to a measurable and controllable direction in the activation space of a large language model. By extracting and manipulating this direction, it is possible to steer model behavior toward or away from a particular writing style without retraining the model.

The results highlight how mechanistic interpretability can provide insights into the internal representations learned by transformer models while enabling practical methods for controllable text generation.